

Microeconomics

Lecture 2: Preferences and Indifference Curves

Marco Rosso

University of Bologna

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Lecture objectives

After this lecture you will be able to:

1. **Understand** the four axioms of rational preferences
2. **Distinguish** ordinal vs cardinal utility
3. **Represent graphically** preferences with indifference curves
4. **Compute** the Marginal Rate of Substitution (MRS)
5. **Apply** partial derivatives to utility functions

Focus of the Lecture

Today we focus ONLY on **what the consumer wants** (preferences).

Next week: **what the consumer can afford** (constraint) and how they find the **optimum**.

The Consumer's Problem

Central question:

How does a rational consumer choose among different alternatives, given limited resources?

To answer, we need to understand:

1. **What the consumer wants** → Preferences
2. **What they can afford** → Budget constraint
3. **How they combine the two** → Optimal choice

Preferences

Today we study only **preferences**: what the consumer likes, how we represent it mathematically, and how we measure the subjective trade-off between goods.

Preferences: Basic Notation

Consumption bundle: a combination of quantities of goods

- Bundle A = (x_A, y_A) : x_A units of good 1, y_A units of good 2
- Bundle B = (x_B, y_B)

Preference relations:

$A \succ B$ "I strictly prefer A to B"

$A \sim B$ "I am indifferent between A and B" (same satisfaction)

$A \succeq B$ "A is at least as good as B" (I prefer A or am indifferent)

Example:

- $A = (4 \text{ pizzas}, 2 \text{ beers})$
- $B = (2 \text{ pizzas}, 3 \text{ beers})$
- If I love pizza: $A \succ B$
- If I seek balance: I might have $A \sim B$

Axioms of Rational Preferences

To model choices mathematically, we assume **rational** preferences:

The Four Axioms

1. **Completeness:** I can always compare two bundles
2. **Transitivity:** preferences are consistent
3. **Monotonicity:** more is better (non-satiation)
4. **Convexity:** I prefer balanced bundles

Note

These are **simplifying assumptions**, not universal truths. They allow us to build tractable mathematical models.

Axiom 1: Completeness

Definition

For every pair of bundles A and B, the consumer is able to compare them.

Formally: $A \succ B$, or $B \succ A$, or $A \sim B$

Interpretation:

- There is no “I don’t know” or “I can’t decide”
- The consumer always knows what they prefer (or whether they are indifferent)
- Even between alternatives never tried

Examples:

- Pizza vs hamburger: I can decide even if I have never tried the hamburger
- Studying vs Netflix: I can compare even if it depends on my mood

Possible violation: “I don’t know which I prefer between economics and engineering”

→ But in the end I *must* choose, so I reveal a preference

Axiom 2: Transitivity

Definition

If I prefer A to B, and B to C, then I prefer A to C.

Formally: If $A \succ B$ and $B \succ C$, then $A \succ C$

Interpretation:

- Preferences are **consistent**
- There are no contradictions or cycles
- Essential for the existence of a utility function

Valid example (transitive):

- Apple $\succ$ Banana
- Banana $\succ$ Orange
- Therefore: Apple $\succ$ Orange

NON-transitive example (irrational):

- Apple $\succ$ Banana
- Banana $\succ$ Orange
- Orange $\succ$ Apple $\rightarrow$ **Cycle! Contradiction!**

Transitivity: Why It Matters

Consequence of a violation: Without transitivity, the consumer can be “manipulated” into an endless cycle of trades.

Example (Money Pump):

Suppose cyclical preferences: Apple $\succ$ Banana $\succ$ Orange $\succ$ Apple

1. You start with 1 Orange
2. I offer: “Trade Orange for Banana + pay €0.10” $\rightarrow$ You accept (Banana $\succ$ Orange)
3. I offer: “Trade Banana for Apple + pay €0.10” $\rightarrow$ You accept (Apple $\succ$ Banana)
4. I offer: “Trade Apple for Orange + pay €0.10” $\rightarrow$ You accept (Orange $\succ$ Apple)
5. **Result:** You are back to the Orange, but you have paid €0.30!
6. I repeat forever $\rightarrow$ you go broke!

Summary

Transitivity protects the consumer from self-defeating behavior.

Axiom 3: Monotonicity (Non-Satiation)

Definition

More is better. If a bundle has at least as much of every good, and more of at least one, then it is preferred.

Formally: If $x_A \geq x_B$ and $y_A \geq y_B$, with at least one strict inequality, then $A \succ B$

Example:

- $A = (3 \text{ pizzas}, 4 \text{ beers})$
- $B = (3 \text{ pizzas}, 2 \text{ beers})$
- $A \succ B$ because it has more beers (for the same number of pizzas)

Economic interpretation:

- There is no complete satiation
- Goods are always desirable

Graphical implication: Indifference curves are downward-sloping!

Monotonicity: Discussion

When is it violated?

1. True satiation:

- “10 ice creams make me sick” → too much of a good reduces utility
- Solution: we work within “reasonable” ranges

2. Bads instead of goods:

- Pollution, noise, commuting time
- Solution: we redefine “reduction of the bad” = good (e.g. clean air instead of pollution)

3. Disposal costs:

- “I don’t want 100 kg of zucchini” → I have to pay for them and dispose of them
- Solution: we distinguish between “acquisition” and “possession”

In Practice

Monotonicity is reasonable for most consumer goods in normal quantities.

Axiom 4: Convexity

Definition

Consumers prefer “balanced” bundles to extreme bundles.

Formally: if $A \sim B$, then any weighted average $(1 - t)A + tB \succeq A$ (with $t \in [0, 1]$)

Example:

- $A = (10 \text{ pizzas}, 0 \text{ beers})$
- $B = (0 \text{ pizzas}, 10 \text{ beers})$
- Suppose $A \sim B$ (same level of satisfaction)
- Then $C = (5 \text{ pizzas}, 5 \text{ beers}) \succeq A$ and $C \succeq B$

Intuition:

- **Variety** is preferred to extreme specialization
- The 10th pizza in a row is worth less than the 1st (*diminishing marginal utility*)
- Diminishing willingness to substitute (we will see this with the MRS)

Graphical implication: Indifference curves **convex to the origin**

Convexity: Discussion

When is it violated?

1. Perfect substitutes:

- Red pencils vs blue pencils $\rightarrow$ I only care about the total
- $(5 \text{ red}, 5 \text{ blue}) \sim (10 \text{ red}, 0 \text{ blue}) \sim (0 \text{ red}, 10 \text{ blue})$
- **Linear preferences** (not strictly convex!)

2. Perfect complements:

- Left vs right shoes $\rightarrow$ needed in pairs (1:1)
- $(10 \text{ right}, 5 \text{ left}) \sim (5 \text{ right}, 5 \text{ left}) \rightarrow$ useless excess
- **“L-shaped” preferences** (not strictly convex!)

3. Extreme tastes:

- “I only love pizza, I hate beer” $\rightarrow$ I prefer $(10 \text{ pizzas}, 0 \text{ beers})$
- Rare but possible

In Practice

Convexity is reasonable for most “normal” goods with imperfect substitutability.

From Preferences to the Utility Function

Idea: Instead of comparing every pair of bundles (infinitely many comparisons!), can we assign a **number** to each bundle?

Definition: Utility Function

A **utility function** $U(x, y)$ assigns a real number to each bundle (x, y) , such that:

$$A \succ B \iff U(A) > U(B)$$

$$A \sim B \iff U(A) = U(B)$$

Interpretation:

- $U(\cdot)$ is a **numerical representation** of preferences
- Bundles with higher utility are preferred
- The absolute number has NO intrinsic meaning (**ordinal** utility)

Example:

- $U(A) = 10, U(B) = 5 \rightarrow$ we only know that $A \succ B$
- We CANNOT say “A is twice as good as B”!

Ordinal vs Cardinal Utility

Ordinal Utility (the one we use!)

- Only the **ranking** matters
- $U(A) = 10, U(B) = 5$
same info as
 $U(A) = 100, U(B) = 2$
- I can apply increasing monotonic transformations
- E.g.: $V = 2U + 3, V = U^2, V = \ln U$

Properties:

- Preserves preferences
- No interpersonal comparisons
- Sufficient for consumer theory

We will always use **ordinal utility**: only the ranking of preferences matters, not absolute values.

Cardinal Utility (we do NOT use it)

- Numbers have absolute meaning
- $U(A) = 10, U(B) = 5$
means “A is twice as good”
- Only affine transformations:
 $V = aU + b$ (with $a > 0$)
- E.g.: Celsius/Fahrenheit scale

Use:

- Theory of choice under uncertainty (expected utility)
- Interpersonal comparisons (welfare)
- Not needed for choices without uncertainty

Monotonic Transformations

Property

If $U(x, y)$ represents preferences, so does $V(x, y) = f(U(x, y))$, provided that f is **strictly increasing**.

Example:

Original function: $U(x, y) = x \cdot y$

Valid transformations (they preserve the ranking):

- $V_1(x, y) = 2U = 2xy$ ✓
- $V_2(x, y) = U + 10 = xy + 10$ ✓
- $V_3(x, y) = \ln U = \ln(xy) = \ln x + \ln y$ ✓
- $V_4(x, y) = U^{1/2} = \sqrt{xy}$ ✓

NON-valid transformation:

- $V_5(x, y) = -U = -xy$ (it reverses the ranking!)

Implication: We can choose the most convenient form for computations!

Three Fundamental Utility Functions

In this course we will mainly use these three types:

1. **Cobb-Douglas** (“standard” preferences)

$$U(x, y) = x^{\alpha} \cdot y^{\beta} \quad \text{or} \quad U(x, y) = \alpha \ln x + \beta \ln y$$

2. **Perfect Substitutes** (1:1 substitutability)

$$U(x, y) = ax + by$$

3. **Perfect Complements** (fixed proportions)

$$U(x, y) = \min\{ax, by\}$$

Type 1: Cobb-Douglas (1)

General Form

$$U(x, y) = x^{\alpha} \cdot y^{\beta} \quad (\alpha, \beta > 0)$$

or the logarithmic version (equivalent!):

$$U(x, y) = \alpha \ln x + \beta \ln y$$

Parameters:

- α, β = “weights” of the goods (relative importance)
- If $\alpha = \beta$: the two goods are equally important
- If $\alpha > \beta$: good x is more important

Type 1: Cobb-Douglas (2)

Examples:

- $U(x, y) = x \cdot y = x^1 \cdot y^1$ (equal weights)
- $U(x, y) = \sqrt{xy} = x^{1/2} \cdot y^{1/2}$ (equivalent to the previous one!)
- $U(x, y) = x^{0.7} \cdot y^{0.3}$ (preference for x)

Properties:

- Represents “standard” preferences (convex, monotonic)
- Widely used for its mathematical simplicity
- Indifference curves = convex hyperbolas

Type 2: Perfect Substitutes

General Form

$$U(x, y) = ax + by \quad (a, b > 0)$$

Interpretation:

- The goods are **perfect substitutes**
- The consumer is willing to trade them at a constant ratio a/b
- Only the “weighted total” matters

Type 2: Perfect Substitutes (2)

Examples:

- $U(x, y) = x + y$ (red vs blue pencils)
- $U(x, y) = 2x + 3y$ (is €2 worth as much as €3? No! But they can be substitution rates)
- Coca-Cola vs Pepsi (for some consumers)

Properties:

- Indifference curves = **parallel straight lines**
- **NOT strictly convex** (flat)
- Constant rate of substitution (constant MRS)

Type 3: Perfect Complements

General Form

$$U(x, y) = \min\{ax, by\} \quad (a, b > 0)$$

Interpretation:

- The goods are **perfect complements**
- They must be consumed in fixed proportions $x : y = b : a$
- An excess of one good does NOT increase utility

Type 3: Perfect Complements (2)

Examples:

- $U(x, y) = \min\{x, y\}$ (left and right shoes, 1:1 ratio)
- $U(x, y) = \min\{2x, y\}$ (coffee and sugar, 1:2 ratio)
- Cars and gasoline (in the short run)

Properties:

- L-shaped indifference curves
- Corner on the line $ax = by$ (optimal proportion)
- Horizontal/vertical segments (useless excess)

Comparison: The Three Types

Type	Function	Indiff. Curves	Example
Cobb-Douglas	$x^\alpha y^\beta$	Convex hyperbolas	Food vs clothing
Perfect substitutes	$ax + by$	Parallel lines	Red vs blue pencils
Perfect complements	$\min\{ax, by\}$	"L" shape	Right vs left shoes

Question: Which preferences are "realistic"?

Answer:

- **Cobb-Douglas:** The most common for "normal" goods (food, clothes, entertainment)
- **Substitutes:** Rare (few goods are perfect substitutes)
- **Complements:** Rare in the extreme form, but useful for some goods (hardware/software, coffee/sugar)

In practice: **Cobb-Douglas is our "benchmark"** for most examples.

Summary

1. Axioms of Rational Preferences

- Completeness $\rightarrow$ I can always compare
- Transitivity $\rightarrow$ consistency (no cycles)
- Monotonicity $\rightarrow$ more is better
- Convexity $\rightarrow$ I prefer balanced bundles

2. Utility Function

- Numerical representation of preferences
- Ordinal (only the ranking matters)
- Monotonic transformations preserve preferences

3. Three Fundamental Types

- Cobb-Douglas: $x^\alpha y^\beta$ (standard)
- Perfect substitutes: $ax + by$
- Perfect complements: $\min\{ax, by\}$

Graphical Representation of Preferences

Problem: Preferences are abstract. How can we visualize them?

Solution: Indifference curves

Definition

An **indifference curve** is the set of all bundles that give the consumer the same level of utility.

Formally: Level curve $\bar{U}$:

$$IC_{\bar{U}} = \{(x, y) : U(x, y) = \bar{U}\}$$

Interpretation:

- All points on the same curve: consumer is **indifferent**
- Moving along the curve: composition changes, utility constant
- Moving across different curves: utility level changes

Properties of Indifference Curves

Four fundamental properties:

1. **Negative slope** (from monotonicity)
2. **They never intersect** (from consistency)
3. **Curves farther from the origin = higher utility** (from monotonicity)
4. **Convex to the origin** (from convexity)

Property 1: Negative Slope (1)

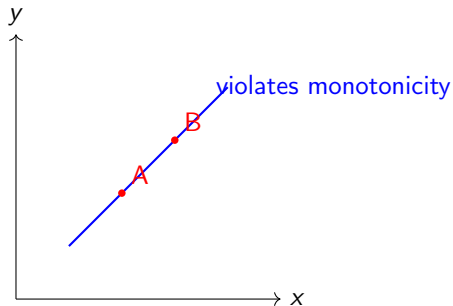
Claim: Indifference curves have a negative slope.

Proof (intuition):

- Start from a point A on the indifference curve
- If I **increase** x (more of good 1) $\rightarrow$ utility rises (monotonicity)
- To stay on the same curve (constant utility), I must **decrease** y
- Therefore: when $x \uparrow$, then $y \downarrow \rightarrow$ negative slope $\checkmark$

Property 1: Negative Slope (2)

Counterexample: A positive slope would violate monotonicity



If slope +, $A \sim B$ but B has more of everything!

Property 2: They Do Not Intersect

Claim: Two indifference curves can never intersect.

Proof by contradiction:

1. Suppose two curves ($U_1 < U_2$) intersect at point A
2. Point B on curve U_2 (different from A): $U(B) = U_2$
3. Point C on curve U_1 (different from A): $U(C) = U_1$
4. From the intersection: $U(A) = U_1$ (because A is on curve 1)
5. But also: $U(A) = U_2$ (because A is on curve 2)
6. Therefore: $U_1 = U_2 \rightarrow$ **Contradiction!**

Conclusion

An intersection would violate the very definition of an indifference curve: each point can belong to only one utility level.

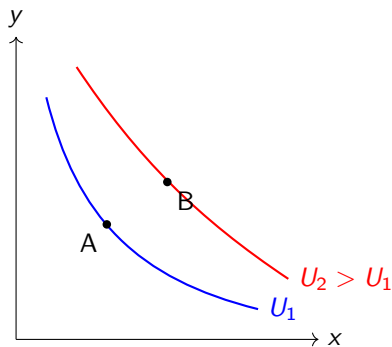
Property 3: Distance from the Origin (1)

Claim: Curves farther from the origin represent higher utility.

Proof:

- Point A on an inner curve (close to the origin)
- Point B on an outer curve (far from the origin)
- By construction: B has more of at least one good (with the same amount of the other, or more of both)
- By monotonicity: $B \succ A$
- Therefore: $U(B) > U(A)$ ✓

Property 3: Distance from the Origin (2)



Property 4: Convexity (1)

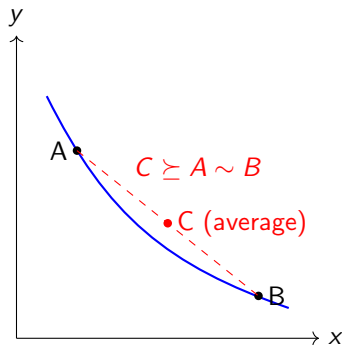
Claim: Indifference curves are convex to the origin (for convex preferences).

Geometric interpretation:

- Take two points A and B on the same indifference curve
- The segment joining them (weighted averages) lies **above** the curve
- That is: “average” bundles give greater or equal utility

Property 4: Convexity (2)

Equivalently: The slope (in absolute value) **decreases** as we move to the right.



Link: This reflects diminishing marginal utility (we will see this with the MRS).

Indifference Map

Definition

An **indifference map** is the set of all of a consumer's indifference curves.

Analogy: Like contour lines on a topographic map

- Each curve = same “altitude” (utility level)
- Higher curves = higher utility
- They do not intersect
- Direction of increasing preference: toward the North-East

Properties:

- Infinitely many curves (one for each level $\bar{U}$)
- They fill the entire positive quadrant $(x, y) \geq 0$
- They **fully** represent the consumer's preferences

Implication: If we know the indifference map, we know all of the consumer's preferences!

Example 1: Cobb-Douglas (1)

Function: $U(x, y) = \sqrt{x \cdot y} = x^{1/2}y^{1/2}$

Equation of the indifference curve at level $\bar{U}$:

$$\sqrt{x \cdot y} = \bar{U} \quad \Rightarrow \quad y = \frac{\bar{U}^2}{x}$$

Features:

- Downward-sloping **hyperbolas**
- Convex to the origin ✓
- Asymptotic to the axes (never touch the axes) $\rightarrow$ I don't want zero of either good
- Higher $\bar{U} \rightarrow$ curve farther out

Example 1: Cobb-Douglas (2)

Numerical points:

- $\bar{U} = 2$: $y = 4/x \rightarrow$ points (1, 4), (2, 2), (4, 1)
- $\bar{U} = 4$: $y = 16/x \rightarrow$ points (2, 8), (4, 4), (8, 2)
- $\bar{U} = 6$: $y = 36/x \rightarrow$ points (4, 9), (6, 6), (9, 4)

Interpretation: “Standard” preferences with imperfect substitutability.

Example 2: Perfect Substitutes (1)

Function: $U(x, y) = x + y$

Equation of the indifference curve at level $\bar{U}$:

$$x + y = \bar{U} \quad \Rightarrow \quad y = \bar{U} - x$$

Features:

- Parallel **straight lines** with slope -1
- **NOT strictly convex** (flat)
- Higher $\bar{U} \rightarrow$ line farther from the origin
- Intercepts: $(0, \bar{U})$ and $(\bar{U}, 0)$

Example 2: Perfect Substitutes (2)

Numerical points:

- $\bar{U} = 5$: $y = 5 - x \rightarrow$ points $(0, 5), (2, 3), (5, 0)$
- $\bar{U} = 10$: $y = 10 - x \rightarrow$ points $(0, 10), (5, 5), (10, 0)$

Interpretation: The consumer is willing to trade x and y at a constant 1:1 ratio. Only the total matters: $(5, 0) \sim (0, 5) \sim (2, 3)$ if $U = 5$.

Example 3: Perfect Complements (1)

Function: $U(x, y) = \min\{x, y\}$

Equation of the indifference curve at level $\bar{U}$:

$$\min\{x, y\} = \bar{U}$$

Segments:

- Vertical segment: $x = \bar{U}$ for $y \geq \bar{U}$ (excess y)
- Horizontal segment: $y = \bar{U}$ for $x \geq \bar{U}$ (excess x)
- Corner: $x = y = \bar{U}$ (optimal 1:1 proportion)

Features:

- “L” shape (right angle)
- Corner on the 45-degree line $x = y$
- NOT differentiable at the corner

Example 3: Perfect Complements (2)

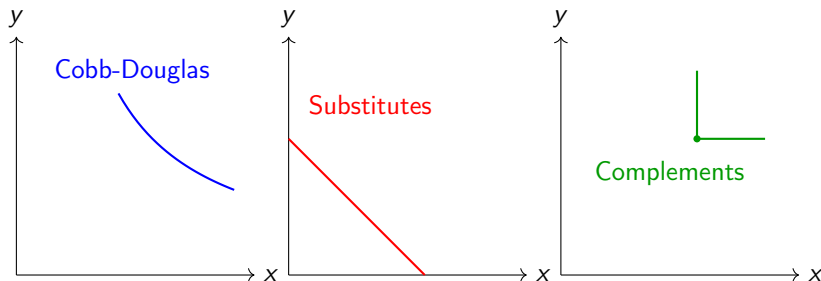
Numerical points:

- $\bar{U} = 3$: L with corner at (3, 3)
- $\bar{U} = 5$: L with corner at (5, 5)
- Note: (5, 10) and (10, 5) both give $U = 5$ (useless excess)

Interpretation: The goods go in pairs. An excess of one does not increase utility.

Graphical Comparison

Indifference curves for $\bar{U} = 2$:



Remarks:

- **Cobb-Douglas:** *diminishing* substitutability (curvature)
- **Substitutes:** *constant* substitutability (straight line)
- **Complements:** no substitutability away from the corner (kink)

Summary

1. Indifference Curves

- Graphical representation of preferences
- Set of bundles with the same utility
- Indifference map = all the curves

2. Four Properties

- Negative slope (monotonicity)
- They do not intersect (consistency)
- Farther out = higher utility (monotonicity)
- Convexity (preference for balanced bundles)

3. Three Graphical Shapes

- Cobb-Douglas $\rightarrow$ convex hyperbolas
- Perfect substitutes $\rightarrow$ parallel lines
- Perfect complements $\rightarrow$ L shape

Measuring the Trade-off

Question: How do we measure how willing a consumer is to substitute one good for the other?

Answer: With the **Marginal Rate of Substitution (MRS)**.

Intuitive Definition

The **MRS** is the quantity of good y the consumer is willing to give up to obtain 1 additional unit of good x , while staying on the same indifference curve (constant utility).

Interpretation:

- It measures the **subjective trade-off** between goods
- How much x is “worth” in terms of y to the consumer
- It depends only on preferences (subjective)
- It varies along the curve (except for perfect substitutes)

Example: $MRS = 2$ means “I am willing to give up 2 beers to get 1 more pizza”

MRS as the Slope of the Curve

Geometric interpretation: The MRS is (the absolute value of) the **slope of the indifference curve**.

Formal Definition (geometric)

$$MRS = -\frac{dy}{dx} \Big|_{U=\text{const}}$$

Note: The minus sign is needed because the slope is negative, but we express the MRS as a positive number

Why does it make sense?

- If I increase x by Δx , I must decrease y by Δy to stay on the curve
- The ratio $-\Delta y/\Delta x$ measures exactly “how much y I give up per unit of x ”
- In the limit: $-dy/dx$

Review: Partial Derivatives

Function of two variables: $U(x, y)$

Partial derivative with respect to x :

$$\frac{\partial U}{\partial x} = \lim_{\Delta x \rightarrow 0} \frac{U(x + \Delta x, y) - U(x, y)}{\Delta x}$$

Interpretation: How much U changes if I increase x slightly, holding y fixed.

Similarly for y :

$$\frac{\partial U}{\partial y} = \lim_{\Delta y \rightarrow 0} \frac{U(x, y + \Delta y) - U(x, y)}{\Delta y}$$

Notation:

- $\frac{\partial U}{\partial x}$ = partial derivative with respect to x
- U_x = shorthand notation (we will use this)
- Similarly: $U_y = \frac{\partial U}{\partial y}$

Marginal Utilities

Definition

The **marginal utility** of good x is the partial derivative of U with respect to x :

$$MU_x = \frac{\partial U}{\partial x}$$

Similarly for y :

$$MU_y = \frac{\partial U}{\partial y}$$

Economic interpretation:

- MU_x = benefit (increase in utility) of consuming 1 more unit of x , holding y fixed
- MU_y = benefit of consuming 1 more unit of y , holding x fixed

Typical property: $MU_x, MU_y > 0$ (monotonicity: more is better)

Diminishing marginal utility: Often $\frac{\partial^2 U}{\partial x^2} < 0 \rightarrow$ The 10th pizza is worth less than the 1st

MRS as a Ratio of Marginal Utilities (1)

Fundamental theorem:

MRS Formula

$$MRS = \frac{MU_x}{MU_y} = \frac{\partial U / \partial x}{\partial U / \partial y}$$

Proof (intuitive): Along the indifference curve: $dU = 0$ (constant utility)

From the total differential formula:

$$dU = MU_x \cdot dx + MU_y \cdot dy$$

MRS as a Ratio of Marginal Utilities

We impose $dU = 0$:

$$MU_x \cdot dx + MU_y \cdot dy = 0$$

$$MU_x \cdot dx = -MU_y \cdot dy$$

$$-\frac{dy}{dx} = \frac{MU_x}{MU_y}$$

Therefore:

$$MRS = -\frac{dy}{dx} \Big|_{U=\text{const}} = \frac{MU_x}{MU_y} \quad \checkmark$$

Economic Interpretation of the MRS

$$MRS = \frac{MU_x}{MU_y}$$

What does it mean?

- **Numerator** (MU_x): How much I gain (in utility) if I consume 1 more unit of x
- **Denominator** (MU_y): How much I gain if I consume 1 more unit of y
- **Ratio**: How many units of y I must give up to offset 1 unit of x

Numerical example:

- $MU_x = 4$ (1 more pizza gives 4 extra “utils”)
- $MU_y = 2$ (1 more beer gives 2 extra “utils”)
- $MRS = 4/2 = 2$

Interpretation: “To me, 1 pizza is worth 2 beers” → I am willing to give up 2 beers to get 1 pizza

Note: The absolute values of MU do not matter (ordinal utility!), only the ratio does.

Computing the MRS: Cobb-Douglas (1)

Function: $U(x, y) = x^\alpha \cdot y^\beta$

Step 1: Compute the marginal utilities

$$MU_x = \frac{\partial U}{\partial x} = \alpha x^{\alpha-1} y^\beta$$

$$MU_y = \frac{\partial U}{\partial y} = \beta x^\alpha y^{\beta-1}$$

Step 2: Compute the MRS

$$MRS = \frac{MU_x}{MU_y} = \frac{\alpha x^{\alpha-1} y^\beta}{\beta x^\alpha y^{\beta-1}} = \frac{\alpha}{\beta} \cdot \frac{y}{x}$$

Computing the MRS: Cobb-Douglas (2)

Special case: $U(x, y) = \sqrt{xy} = x^{1/2}y^{1/2} \rightarrow \alpha = \beta = 1/2$

$$MRS = \frac{1/2}{1/2} \cdot \frac{y}{x} = \frac{y}{x}$$

Numerical example: At the point $(x, y) = (2, 8)$:

$$MRS = \frac{8}{2} = 4$$

Interpretation: At that point, the consumer is willing to give up 4 beers for 1 pizza.

Computing the MRS: Perfect Substitutes (1)

Function: $U(x, y) = ax + by$

Step 1: Compute the marginal utilities

$$MU_x = \frac{\partial U}{\partial x} = a$$

$$MU_y = \frac{\partial U}{\partial y} = b$$

Step 2: Compute the MRS

$$MRS = \frac{MU_x}{MU_y} = \frac{a}{b} \quad (\text{constant!})$$

Computing the MRS: Perfect Substitutes (2)

Example: $U(x, y) = x + y \rightarrow a = b = 1$

$$MRS = \frac{1}{1} = 1 \quad \text{everywhere}$$

Interpretation:

- The rate of substitution is **constant** (it does not depend on (x, y))
- The consumer is always willing to trade at the ratio $a : b$
- Consistent with linear indifference curves (constant slope)

Computing the MRS: Perfect Complements

Function: $U(x, y) = \min\{x, y\}$

Problem: The min function is not differentiable at the “kink” (corner)!

Case-by-case analysis:

1. **Vertical segment** ($y > x$):

Increasing y does not increase utility (U depends only on x) $\rightarrow MU_y = 0$, so $MRS = MU_x/0 = \infty$

2. **Horizontal segment** ($x > y$):

Increasing x does not increase utility (U depends only on y) $\rightarrow MU_x = 0$, so $MRS = 0/MU_y = 0$

3. **Corner** ($x = y$):

MRS **not defined** (curve not differentiable at the kink)

Interpretation:

- Away from the corner: no substitutability ($MRS = 0$ or ∞)
- At the corner: perfect proportion, any trade reduces utility

Diminishing MRS

Empirical observation: For convex preferences (Cobb-Douglas), the MRS **decreases** along the indifference curve (moving to the right).

Why? Diminishing marginal utility:

- Point A: (1 pizza, 10 beers) $\rightarrow$ I have little x , a lot of y
 MU_x high (I value pizza highly), MU_y low (plenty of beer) $\rightarrow MRS = MU_x/MU_y$ **high**
- Point B: (5 pizzas, 2 beers) $\rightarrow$ I have a lot of x , little y
 MU_x low (plenty of pizza), MU_y high (I value beer highly) $\rightarrow MRS = MU_x/MU_y$ **low**

Numerical example (Cobb-Douglas $MRS = y/x$):

- A = (1, 10): $MRS = 10/1 = 10$ (I would give 10 beers for 1 pizza)
- B = (5, 2): $MRS = 2/5 = 0.4$ (I would give 0.4 beers for 1 pizza)

Link: Diminishing MRS $\leftrightarrow$ convex curves $\leftrightarrow$ diminishing marginal utility

Exercise 1: Computing the MRS

Utility function: $U(x, y) = 2x^{1/2} + y$

Question: Compute the MRS at the point $(x, y) = (4, 5)$

Solution:

Step 1: Marginal utilities

$$MU_x = \frac{\partial U}{\partial x} = 2 \cdot \frac{1}{2} x^{-1/2} = \frac{1}{\sqrt{x}}$$

$$MU_y = \frac{\partial U}{\partial y} = 1$$

Step 2: General MRS

$$MRS = \frac{MU_x}{MU_y} = \frac{1/\sqrt{x}}{1} = \frac{1}{\sqrt{x}}$$

Step 3: Evaluate at $(4, 5)$

$$MRS(4, 5) = \frac{1}{\sqrt{4}} = \frac{1}{2}$$

Interpretation: At the point $(4, 5)$, the consumer is willing to give up 0.5 units of y to obtain 1 unit of x .

Exercise 2: MRS and Convexity

Function: $U(x, y) = x^{0.6}y^{0.4}$

Question 1: Compute the MRS as a function of x and y .

Solution:

$$MU_x = 0.6x^{-0.4}y^{0.4}, \quad MU_y = 0.4x^{0.6}y^{-0.6}$$

$$MRS = \frac{0.6x^{-0.4}y^{0.4}}{0.4x^{0.6}y^{-0.6}} = \frac{0.6}{0.4} \cdot \frac{y}{x} = 1.5 \cdot \frac{y}{x}$$

Question 2: Verify that the MRS is diminishing along the curve.

Solution: $MRS = 1.5 \cdot y/x$

- Along an indifference curve: when $x \uparrow$, then $y \downarrow$ (negative slope)
- Therefore: $y/x \downarrow \rightarrow MRS \downarrow \checkmark$

Conclusion: Preferences are convex (diminishing MRS).

Exercise 3: Comparing MRS (1)

Two consumers with different preferences:

- Consumer A: $U_A(x, y) = x \cdot y$
- Consumer B: $U_B(x, y) = 2x + y$

Question: Who has the higher MRS at the point (2, 4)?

Solution:

Consumer A (Cobb-Douglas):

$$MRS_A = \frac{y}{x} = \frac{4}{2} = 2$$

Consumer B (perfect substitutes):

$$MRS_B = \frac{2}{1} = 2$$

Answer: At this point, both have the same $MRS=2$!

Exercise 3: Comparing MRS (2)

Key difference:

- For A, the MRS **depends on the point**: along the same indifference curve, it decreases as x grows (diminishing substitutability).
- For B, the MRS is **constant everywhere** ($=2$), regardless of the point chosen.

Counterexample: If they move to (4,2) (on A's same indifference curve):

- A: $MRS_A = 2/4 = 0.5$ (now willing to give up less y for x)
- B: $MRS_B = 2/1 = 2$ (never changes)

Exercise 4: Drawing the Curve from the MRS

Given: $MRS = 3$, constant everywhere.

Question 1: What type of preferences?

Answer: Perfect substitutes (constant MRS)

Question 2: Write a possible utility function.

Answer: $U(x, y) = 3x + y$ (Check: $MU_x = 3$, $MU_y = 1$, so $MRS = 3/1 = 3$ ✓)

Question 3: Draw the indifference curve for $U = 6$.

Answer: $3x + y = 6 \rightarrow y = 6 - 3x$

Straight line with intercepts $(0, 6)$ and $(2, 0)$, slope -3 .

Exercise 5: Numerical Check

Function: $U(x, y) = \ln x + 2 \ln y$

Question: Compute the MRS at (1, 4) and (4, 2) (same indifference curve). Is it diminishing?

Solution:

$$MU_x = \frac{1}{x}, \quad MU_y = \frac{2}{y}$$
$$MRS = \frac{MU_x}{MU_y} = \frac{1/x}{2/y} = \frac{y}{2x}$$

Point (1, 4):

$$MRS(1, 4) = \frac{4}{2 \cdot 1} = 2$$

Point (4, 2):

$$MRS(4, 2) = \frac{2}{2 \cdot 4} = 0.25$$

Answer: Yes, the MRS decreases (from 2 to 0.25) moving to the right along the curve → convex preferences ✓

Overall Summary

Part 1: Axioms and Utility Functions

- 4 rationality axioms (completeness, transitivity, monotonicity, convexity)
- Ordinal utility (only the ranking matters)
- 3 types: Cobb-Douglas, substitutes, complements

Part 2: Indifference Curves

- Graphical representation of preferences
- 4 properties (negative slope, non-intersecting, distance, convexity)
- Graphical shapes: hyperbolas, straight lines, “L”

Part 3: Marginal Rate of Substitution

- $MRS = \text{slope of the curve} = MU_x/MU_y$
- Measures the subjective trade-off between goods
- Diminishing MRS $\leftrightarrow$ convexity
- Computation for different functions

What we should be able to do after this lecture

- ✓ Recognize whether preferences satisfy the axioms
- ✓ Write utility functions for different types of preferences
- ✓ Draw indifference curves
- ✓ Compute the MRS given $U(x, y)$
- ✓ Interpret the MRS economically
- ✓ Check convexity (diminishing MRS)

Optional Exercises for Home

1. Given $U(x, y) = x^{0.3}y^{0.7}$, compute the MRS at $(9, 4)$.
2. Given $U(x, y) = 3x + 5y$, what is the MRS? What type of preferences?
3. If $MRS = 4$ at the point $(2, y)$ and $U(x, y) = \sqrt{xy}$, find y .
4. Verify that $U(x, y) = xy$ and $V(x, y) = 2xy + 10$ represent the same preferences.
5. Sketch qualitatively the indifference curves for:
 - a) $U = x + 2y$
 - b) $U = \min\{2x, y\}$
 - c) $U = x^{0.5}y^{0.5}$